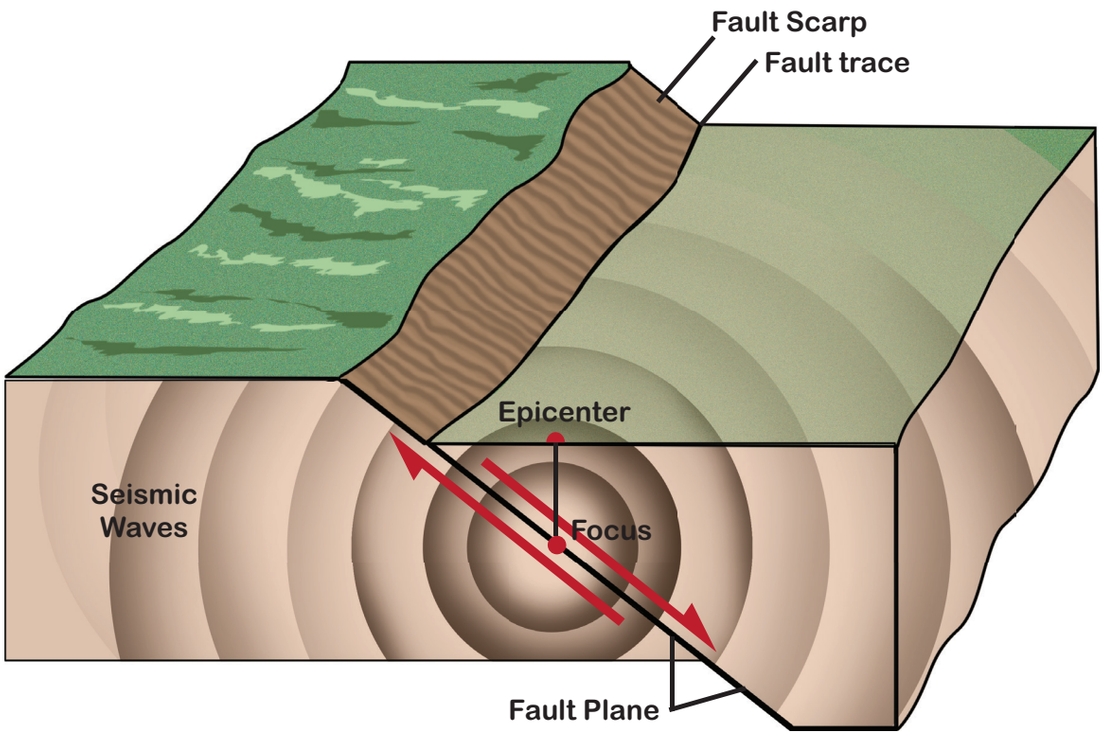



**Figure 3.22** Parts of an Earthquake

a. ***P or Primary or Compressional waves*** are the fastest seismic waves (6 km/ sec. in the upper crust). They cause the matter to oscillate forward and backward, parallel to the motion of the seismic wave front. P waves push (compress) and pull (dilate) the rock that they pass through. They pass through all medium.

b. ***S or Secondary or Shear waves*** are slower than the primary waves (3.5 km/sec. in the upper crust). They cause matter to oscillate side to side, perpendicular to the motion of the wave front. S waves shear the rock that they pass through. They pass through only solid medium.

II. ***Surface Waves*** are the waves that travel along the earth's surface. They are slower than body waves. They cause damage during earthquakes.

***Love waves*** shake the ground side to side like S wave.

***Rayleigh waves*** displace the ground like rolling ocean waves. The ground rolls forward and up and then down and backwards. This is similar to a p wave but with the extra up-down motion.

### 3.9.1 Measuring the earthquake

It is estimated that about 100,000 earthquakes occur but all cannot be felt. A few earthquakes may be severe causing huge damage to property. Earthquake magnitude is measured on the Richter scale (named after the seismologist who devised it), which rates them on a scale of 1 to 10. Earthquake intensity is measured on the modified Mercalli scale, which ranges from 1 to 12, depending upon the intensity. The seismograph is an instrument used to detect and record seismic waves created by the earthquakes.



TN_Govt_Geography_XI_Std Unit_03.indd 65 2/3/2020 4:22:50 PM